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DAY 4

Task 2

AI Models for Virtual Try-on Research

Goal

This research aimed at exploring AI-based technologies used in Virtual Try-On applications, which let people see themselves wearing different outfits digitally, without actually wearing them. This research considered the usage of AI, computer vision, and deep learning techniques in current applications in fashion technology.

Virtual Try-On Systems

Virtual Try-On (VTO) systems are AI-based solutions allowing users to put on clothes digitally and try them using either a virtual avatar or live camera input. It superimposes clothes over the user's body, taking into account several important aspects, including: Alignment of the body, Fit of the cloth, Pose accuracy, Realism of the outcome

These solutions are used in many places, including:

- E-commerce platforms
- Fashion apps
- Augmented Reality/VR fashion technology
- Key Elements of Virtual Try-On Technology

1. Human Pose Detection

Human pose detection detects body parts such as: Shoulders, Arms, Waist, Legs

The function is to ensure proper fitting of the clothes.

Techniques Employed: MediaPipe, OpenPose

2. Body Segmentation

Body segmentation segregates the human body from the surrounding environment.

Benefits: Accurate placement of clothes, Background elimination, Improved visibility of clothes.

- Techniques Employed:
- Semantic Segmentation
- Selfie Segmentation
- Human Parsing

3. Cloth Warping

Cloth warping deforms the cloth based on body size and pose.

Functions: Natural stretching of clothes,, Maintenance of cloth quality, Alignment of sleeves and borders

4. Image Rendering

In this step, the system performs: Combination of user image, Body landmarks, Clothing image

Popular AI Models for Virtual Try-On

1. VITON (Virtual Try-On Network)

Features: Generates lifelike clothing overlays, Utilizes image synthesis methods, Preserves body pose

Strengths: High-quality outputs, Fits well to the real-life body

Weaknesses: Needs high computational resources

2. CP-VTON

Features: Better cloth warping technique, Improvement of clothing overlay, Better texture conservation

Strengths: Higher cloth fitting accuracy, Higher visual realism

Weaknesses: Hard to implement

3. TryOnGAN

Features: Based on GAN (generative adversarial networks), Provides highly realistic try-on photos

Strengths: Better image quality, Cloth looks more natural

Weaknesses: Requires powerful GPU

Virtual Try-On AI Workflow

Step 1 – Capture User Image

The algorithm collects: Body image, Live webcam stream, Pose data

Step 2 – Identify Body Landmarks

AI models identify: Shoulder landmarks, Angle of arms, Body waistline, Overall body posture

Step 3 – Separate Human Body

Human body is isolated to place an overlay accurately.

Step 4 – Overlay Selected Clothing Items

Clothing items are: Adjusted for size, Warped, Correctly placed

Step 5 – Render Final Results

Benefits of Using Virtual Try-On AI Models

Benefit	Description
Improved Customer Experience	Customers will be able to try out virtual outfits
Decreased Return Rate	Better size approximation
Customized Fashion	Provides recommendations for outfits
Saves Time	No physical fitting is required
Engagement	Greater customer engagement

Obstacles in the Development of Virtual Try-On AI

Obstacle	Description
Complicated Body Poses	Difficult alignment between body and clothes
Lighting Differences	Image quality
Deformations of Clothes	Hard to keep natural fabric deformations
Real-Time Computation	Optimization
Occultations	Human body obstructs clothes

Results of Scientific Research

The scientific work gave a thorough knowledge about:

- Fashion technologies based on Artificial Intelligence
- Methods of Human Pose Estimation
- Segmentation of the Human Body
- Warping of Clothes
- Image generation using deep learning

The research also highlighted the use cases of Virtual Try-On AI systems in future AI/ML applications such as:

- Smart Fashion Assistants
- Online Clothes Shops
- Outfit Recommendation Platforms

Conclusion

The Virtual Try-On AI utilizes Computer Vision, Deep Learning, and Pose Detection for creating real-life digital clothes experiences. Modern virtual try-on models such as VITON, CP-VTON, and TryOnGAN prove the potential of applying Artificial Intelligence to modern fashion and e-commerce.